MATH 127 Calculus for the Sciences

Lecture 21

Today's lecture

Last time
IVT, discontinuities.

This time
Course note coverage Section 3.1.5 and 3.2.1
Limit at infinity
Indetermi~~nant~~ nate form

Limit at infinity

We have seen taking a limit of a function as x goes to a for some number a:

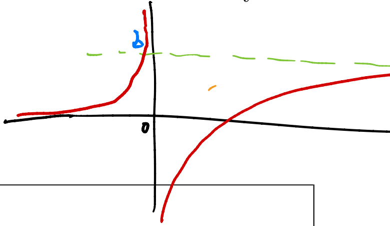
$$\lim_{x \to a} f(x).$$

This gives you the behaviour of $f(x)$ near $x = a$.

You can also take limit to infinity, which tells the behaviour at infinity. You might know this as horizontal asymptotes:

$$\lim_{x \to \infty} f(x),$$

$$\lim_{x \to -\infty} f(x).$$



Remark What are $f(\infty)$ and $f(-\infty)$?

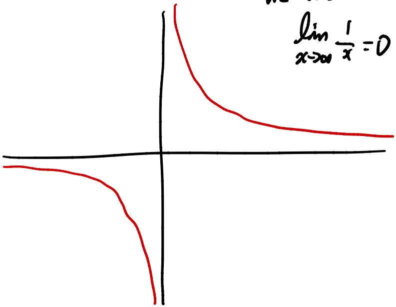
Limit at infinity

Example What is

$$\lim_{x\to\infty} \frac{1}{x}$$

Method 1

From the graph we see $\lim_{x\to\infty} \frac{1}{x} = 0$



Method 2

As x approaches $+\infty$.

the denominator of $\frac{1}{x}$ is bigger and bigger.

So $\frac{1}{x}$ is closer and closer to 0

Hence $\lim_{x\to\infty} \frac{1}{x} = 0$.

Method 3

$\lim_{x\to\infty} \frac{1}{x}$ is the determinate form $\frac{1}{\infty}$

so $\lim_{x\to\infty} \frac{1}{x} = 0$

Limit at infinity

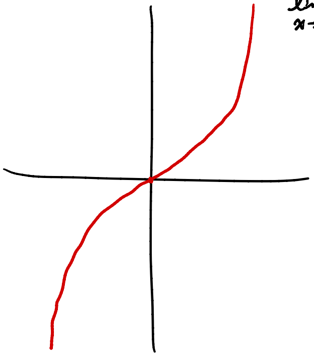
Example What is

$$\lim_{x \to -\infty} x^3$$

Method 1

By the graph, we know

$$\lim_{x \to -\infty} x^3 = -\infty$$



Method 2

As x approaches $-\infty$, ie.

x is a bigger and bigger negative number

The cube of x is also big and negative.

Hence $\lim_{x \to -\infty} x^3 = -\infty$.

Limit at infinity

Example What is

$$\lim_{x\to\infty} e^{1/x}/x$$

Step 1: limit of $\frac{1}{x}$

As $x \to \infty$ $\frac{1}{x}$ approaches 0 from the positive side.

Step 2: limit of $e^{1/x}$

As $\frac{1}{x} \to 0^+$. the exponent of $e^{\frac{1}{x}}$ is a positive number close to 0.

So the limit of $e^{\frac{1}{x}}$ is 1 from the positive side.

Step 3: limit of $e^{1/x}/x$

As $e^{1/x} \to 1^+$, and $x \to \infty$, the denominator of $\frac{e^{1/x}}{x}$ is big

while the numerator is close to 1 so

the quotient approaches 0. Hence $\lim_{x\to\infty} e^{1/x}/x = 0$

Limit at infinity

Example What is
$$\lim_{x\to\infty} \frac{2x^3+x}{x^3+1}$$

$$\lim_{x\to\infty} \frac{2x^3+x}{x^3+1} = \lim_{x\to\infty} \frac{\left(\frac{2x^3+x}{x^3}\right)}{\left(\frac{x^3+1}{x^3}\right)}$$

$$= \lim_{x\to\infty} \frac{\left(2+\frac{1}{x^2}\right)}{\left(1+\frac{1}{x^3}\right)}$$

$$= \frac{\lim_{x\to\infty} 2+\frac{1}{x^2}}{\lim_{x\to\infty} 1+\frac{1}{x^3}}$$

$$= \frac{2+\lim_{x\to\infty}\frac{1}{x^2}}{1+\lim_{x\to\infty}\frac{1}{x^2}} = 2$$

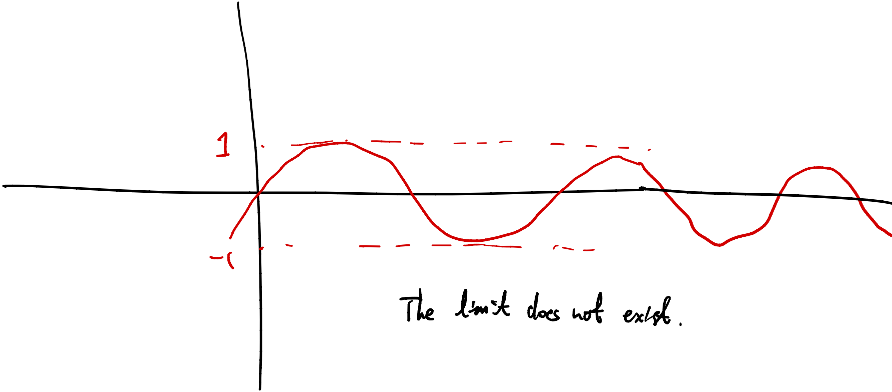
$$= \frac{\lim_{x\to\infty} 2x^3+x}{\lim_{x\to\infty} x^3+1}$$

$$= \frac{\infty}{\infty}$$ incorrect.

Limit at infinity

Example What is

$$\lim_{x\to\infty} \sin(x)$$



The limit does not exist.

Determinate form

Recall we could have a determinate form

$$\lim_{x \to 0^+} \frac{1}{x} \rightsquigarrow \frac{1}{0^+}$$

This is not saying the right hand side is $\frac{1}{0}$, because that doesn't make sense. It is a notation to denote that we are dividing a constant number by a super small positive number, so the result *must be* ∞.

This is called a **determinate form**. Another determinate form is $\frac{0}{\infty}$ which yields 0 because you divide a small number by a big positive number.

Question Can you come up with a determinate form that yields $-\infty$?

$\frac{-1}{0^+}$, $\frac{1}{0^-}$, $\frac{+\infty}{0^-}$. ∞^∞, 0^∞

Indeterminate form

We could also have

$$\lim_{x\to-\infty} \frac{x+1}{x} \rightsquigarrow \frac{-\infty}{-\infty}$$

Again, this is not saying the right hand side is $\frac{-\infty}{-\infty}$, because ∞ is not a number and doesn't make sense to divide them. It says we are dividing a big negative number by a big negative number. **But** we do not know which one is bigger:

- If the numerator is 2 times bigger than the denominator, then the result is 2.
- If the numerator is half the denominator, then the result is $\frac{1}{2}$.

This is called a **indeterminate form** because we can not conclude what the answer must be. Another indeterminate form is $\frac{0^+}{0^+}$.

Question Can you come up with another indeterminate form?

$\frac{\infty}{\infty}$. $\frac{0^-}{0^-}$, ∞^0. 0^0

Indeterminate form

Example Which one of the following is determinate, and which one is indeterminate?

to ∞

- $\lim\limits_{x\to\infty} \dfrac{\ln x}{x}$ is represented by $\dfrac{\infty}{\infty}$, so indeterminate.
- $\lim\limits_{x\to 0^+} \dfrac{\ln x}{x}$ is represented by $\dfrac{-\infty}{0^+}$. so determinate which gives $-\infty$
- $\lim\limits_{x\to\infty} \dfrac{e^x}{x^2+1}$ is represented by $\dfrac{\infty}{\infty}$, so indeterminate.
- $\lim\limits_{x\to 0} \dfrac{\sin(x)}{2x}$ is represented by $\dfrac{0}{0}$. so indeterminate
- $\lim\limits_{x\to 1^-} \dfrac{e^x}{x-1}$ is represented by $\dfrac{e}{0^-}$, so determinate, and gives $-\infty$